

# SparkCalc

SparkCalc computes declared expressions on the live Sparkplug estate and publishes results as their own identity, so a derived value is never mistaken for a measured one.

On the benchIgnition 8.1.19+ and 8.3Runs on: Host

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## 1. Overview

SparkCalc computes declared expressions on the live Sparkplug estate and publishes results as Sparkplug metrics under their own identity (`Calc-<group>`), so a derived value is never mistaken for a measured one. It is aimed at rates, totals, efficiencies, rollups and unit conversions that today live in a PLC, an Ignition expression tag, or a notebook, none of which make the derived value visible to every non-Ignition consumer of the namespace.

SparkCalc is on the bench, not customer ready, per the module's own README; what is proven and what is owed is documented there.

## 2. What You Need

- A SparkBridge Host gateway.

## 3. Install

SparkCalc comes with the Host license. Install it through Config > Modules > Install or Upgrade a Module; no build step is required.

## 4. License

Included with SparkBridge Host. Covered by the suite license key described in the shared install steps: file at `<data>/sparkbridge-license.key`, re-checked every 30 seconds.

## 5. Configure

Declared expressions, staleness budgets, and identity settings are documented in the module's own README and in `docs/LIVEPROOF.md`. Only what is confirmed there is listed below; see the README in the bundle for the full property list.

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Result identity</b>      | Results publish under a <code>calc-&lt;group&gt;</code> identity, with its own edge node and <code>bdSeq</code> lifecycle, distinct from the plant's own identity. The module refuses to produce an identity that could be the plant's.                                                                                                                                                                                         |
| <b>Staleness budgets</b>    | Every declared input has its own staleness budget; there is no single global staleness setting.                                                                                                                                                                                                                                                                                                                                 |
| <b>Bad-quality behavior</b> | If any input is stale, dead, or never seen, the metric evaluates to nothing: bad quality with no value at all, never zero, never NaN, never the last known value.                                                                                                                                                                                                                                                               |
| <b>Provenance</b>           | Every metric's <code>NBIRTH</code> carries <code>calc.expression</code> , <code>calc.inputs</code> , <code>calc.stalenessBudgetsMs</code> , <code>calc.definitionHash</code> , <code>calc.trigger</code> and <code>calc.engine</code> . Every <code>NData</code> carries <code>Quality</code> , <code>calc.quality</code> , <code>calc.stalestInput</code> , <code>calc.ageMs</code> , and when bad, <code>calc.reason</code> . |

## 6. Operate

A computed metric's timestamp is always the stalest contributing input's source timestamp, never the evaluation clock: a ratio of a flow read at 12:00:00 and a level read at 11:59:55 describes the plant as of 11:59:55. When any declared input goes stale, dead, or was never seen, the metric publishes bad quality with no value at all rather than a stale or default value. Because the calculation engine is a pure function of its inputs and its own threaded state, a derived series can be recomputed over history, not only observed forward from now.

## 7. Verify

Confirm published metrics appear under the `calc-<group>` edge node rather than under the plant's own edge, and that the reported timestamp is the stalest contributing input's source timestamp, not the evaluation time. Full proof steps are in `docs/LIVEPROOF.md` in the repo.

## 8. Troubleshoot

| Symptom                                          | Cause                                                                                     | Fix                                                                                                                                   |
|--------------------------------------------------|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Metric never publishes a value                   | One or more declared inputs are stale, dead, or never seen                                | Check the input's own staleness budget and confirm it is publishing on the expected schedule; SparkCalc will not substitute a default |
| Computed value appears under the wrong edge node | Should not occur: the module refuses any identity that could collide with the plant's own | If seen, file it as a defect; this is a designed invariant, not a configuration option                                                |
| Timestamp on a computed metric looks wrong       | Normal: the timestamp is the stalest contributing input's timestamp, not now              | Check <code>calc.stalestInput</code> and <code>calc.ageMs</code> on the metric to see which input is holding it back                  |

## 9. Reference

- Result identity prefix: `calc-<group>`.
- Full config key list and expression surface: module README and `docs/LIVEPROOF.md` in the bundle.

## 10. Support

